

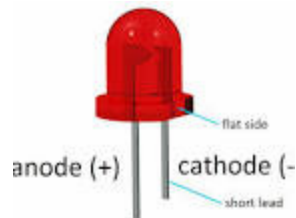
Practical 1

Aim: Displaying different LED patterns with Raspberry Pi.

Hardware Requirements:

- Raspberry Pi
- Ethernet Cable/ Wi-Fi
- Jumper Wires (Female to Female)
- LED

Diagram:



LED Connection:

1. In LED

Code:

```
import RPi.GPIO as GPIO
```

```
import time
```

```
# Get user input
```

```
numTimes = int(input("Enter total number of times to blink: "))
```

```
speed = float(input("Enter length of each blink (seconds): "))
```

```
GPIO.setwarnings(False)
```

```
GPIO.setmode(GPIO.BOARD) # Using physical pin numbers
```

```
# Setup GPIO pins for output
```

```
GPIO.setup(11, GPIO.OUT) # LED1
```

```
GPIO.setup(13, GPIO.OUT) # LED2
```

```
def Blink(numTimes, speed):
```

```
for i in range(numTimes):  
    print("Iteration", i + 1)  
  
    GPIO.output(11, True) # LED1 ON  
    GPIO.output(13, True) # LED2 ON  
    time.sleep(speed)  
  
    GPIO.output(11, False) # LED1 OFF  
    GPIO.output(13, False) # LED2 OFF  
    time.sleep(speed)  
Blink(numTimes, speed)  
GPIO.cleanup()  
print("Done")
```